# Seewoo Lee

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## Education

**University of California Berkeley** — Berkeley
Ph.D in Mathematics — 2018 – 2026 (expected)

- On leave for military service (2019 Fall - 2022 Summer)
- Advisor: Sug Woo Shin

**Pohang University of Science and Technology (POSTECH)** — Pohang
M.S in Mathematics — 2017 – 2018

- Thesis: *Maass wave forms, quantum modular forms and Hecke operators*
- Advisor: YoungJu Choie

**Pohang University of Science and Technology (POSTECH)** — Pohang
B.S. in Mathematics — 2013 – 2017

- *Summa Cum Laude* with top honours in mathematics
- Honor's thesis: *Quantum modular forms and Hecke operators*

## Experiences

**Axiom Math** — Palo Alto
Internship — 2026.01 – present

- Machine Learning & AI for Mathematical Discovery and Reasoning

**CryptoLab** — Seoul
Research Engineer — 2021.05 – 2022.07

- Research on Homomorphic Encryption and application in Machine Learning

**Riiid!** — Seoul
Research Scientist — 2019.07 – 2021.05

- Research on Knowlege Tracing, Score Prediction, Student Dropout Prediction, Item Recommendation

## Research Interests

- Automorphic Forms and Representations, Computational Number Theory, Relative Langlands Program
- Machine Learning and Deep Learning, Formalization of Mathematics, Homomorphic Encryption

## Publications

- Math

1. K. Lee, **S. Lee**, *Machines Learn Number Fields, But How? The Case of Galois Groups*, To appear in Research in Mathematical Sciences, arXiv:2508.06670
2. **S. Lee**, *Shanks bias in function fields*, To appear in Journal de Théorie des Nombres de Bordeaux, arXiv:2509.16142
3. J. Baek, **S. Lee**, *An equilateral triangle of side $> n$ cannot be covered by $n^2 + 1$ unit equilateral triangles homothetic to it*, American Mathematical Monthly, 1-9 (2024)
4. D. Choi, **S. Lee**, *Non-archimedean Sendov's conjecture*, $p$-adic numbers, Ultrametric Analysis and Applications 14, 77-80 (2022)
5. **S. Lee**, *Maass wave forms, Quantum Modular Forms and Hecke Operators*, Res. Mathematical Science 6, 7 (2018), Modular Forms are Everywhere: Celebration of Don Zagier's 65th Birthday
6. **S. Lee**, *Quantum Modular Forms and Hecke Operators*, Res. Number Theory 4, 18 (2018)
7. Y. Chen, R. Chernov, M. Flores, M. F. Bourque, **S. Lee**, B. Yang, *Toy Teichmüller spaces of real dimension 2: the pentagon and the punctured triangle*, Geom. Dedicata 197 (2018), 193-227

- Others

1. F. Lin, K. Nagel, **S. Lee**, J. Jiang, G. Yang, P. Chang, S. Li, N. Sheu, *An Analysis of Silk Density in Spider Webs*, Royal Society Open Science 12: 250455 (2025)
2. **S. Lee**, G. Lee, J. Kim, J. Shin, M. Lee, *HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption*, International Conference on Machine Learning. 2023 (Oral, 155/6538)
3. **S. Lee**, J. Kim, *Revisiting the Convergence Theorem for Competitive Bidding in Common Value Actions*, Economic Theory Bulletin 10, 293-302 (2022)
4. **S. Lee**, K. Kim, J. Shin, J. Park, *Tracing Knowledge for Tracing Dropouts: Multi-Task Training for Study Session Dropout Prediction*, Educational Data Mining. 2021
5. M. Kim, Y. Shim, **S. Lee**, H. Loh, J. Park, *Behavioral Testing of Deep Knowledge Tracing Models*, Educational Data Mining 2021
6. H. Loh, D. Shin, **S. Lee**, J. Baek, C. Hwang, Y. Lee, Y. Cha, S. Kwon, J. Park and Y. Choi, *Recommendation for Effective Standardized Exam Preparation*, LAK21: 11th International Learning Analytics and Knowledge Conference. 2021
7. D. Shin, Y. Shim, H. Yu, **S. Lee**, B. Kim, Y. Choi, *SAINT+: Integrating Temporal Features for EdNet Correctness Prediction*, LAK21: 11th International Learning Analytics and Knowledge Conference. 2021
8. Y. Choi, Y. Lee, D. Shin, J. Cho, S. Park, **S. Lee**, J. Baek, C. Bae, B. Kim, J. Heo, *EdNet: A Large-Scale Hierarchical Dataset in Education*, International Conference on Artificial Intelligence in Education (2021), 69-73
9. J. Kim, **S. Lee**, *Joint Liability and Stochastic Shapley Value*, International Review of Law & Economics 60 (2019), 1-8

## Preprints

1. G. Bates, R. Jesubalan, **S. Lee**, J. Lu, H. Shim, *Ties in function field prime race*, arXiv:2603.21005, submitted
2. **S. Lee**, *Inequalities involving polynomials and quasimodular forms*, arXiv:2602.10536, submitted
3. E. Chen, K. Lau, **S. Lee**, K. Ono, J. Zhang, *Almost all primes are partially regular*, arXiv:2602.05090, submitted
4. G. Bates, R. Jesubalan, **S. Lee**, J. Lu, H. Shim, *Powerful Fibonacci polynomials over finite fields*, arXiv:2601.02664, submitted
5. J. Getz, A. G. Terradillos, F. Hosseinijafari, B. Hu, **S. Lee**, A. Slipper, M.-H. Tomé, H. Yao, A. Zhao, *Modulation groups*, arXiv:2510.23932, submitted
6. J. Baek, **S. Lee**, *Formalizing Mason–Stothers Theorem and its Corollaries in Lean 4*. arXiv:2408.15180
7. **S. Lee**, *Algebraic proof of modular form inequalities for optimal sphere packings*. arXiv:2406.14659, submitted
8. **S. Lee**, Y. Choi, J. Park, B. Kim, J. Shin, *Consistency and Monotonicity Regularization for Neural Knowledge Tracing*, arXiv:2105.00607
9. Y. Choi, Y. Lee, J. Cho, J. Baek, D. Shin, H. Yu, Y. Shim, **S. Lee**, J. Shin, C. Bae, B. Kim, J. Heo, *Assessment Modeling: Fundamental Pre-training Tasks for Interactive Educational Systems*, arXiv:2022.05505

## Awards, Grants & Honours

Department of Mathematics Summer Grant, UC Berkeley . . . . . 2024 Summer
Outstanding Graduate Student Instructor Award, UC Berkeley . . . . . 2024 Spring
Graduate Student Researcher, UC Berkeley . . . . . 2023 Spring, Summer
Kwanjeong Educational Foundation Scholarship, KEF . . . . . 2017–2018
Excellency Award (Top Honours), Dept. of Mathematics, POSTECH . . . . . 2017
POSTECH Outstanding Talent Development Scholarship, POSTECH . . . . . 2013–2016
National Science and Technology Scholarship, KOSAF . . . . . 2013–2016
Silver medals, Undergraduate Mathematical Competition, KMS . . . . . 2013, 2015, 2016
31st place, ACM-ICPC Daejeon Regional, ACM . . . . . 2015
Grand prize, POSTECH Programming Contest, Dept. of Computer Science, POSTECH . . . . . 2015
Honorable mention, Korean Olympiad of Informatics, NIA . . . . . 2012

## Teaching Experience

**Graduate Student Instructor (T.A.)** Berkeley
UC Berkeley 2019 – Present

- (2026 Spring) Multivariable Calculus
- (2025 Fall) Introduction to Abstract Algebra
- (2025 Spring) Cryptography
- (2025 Spring) Introduction to Mathematical Logic

- (2024 Fall) Abstract Linear Algebra
- (2024 Spring) Methods of Mathematics: Calculus, Statistics, and Combinatorics
- (2023 Fall) Methods of Mathematics: Calculus, Statistics, and Combinatorics
- (2022 Fall) Multivariable Calculus
- (2019 Spring) Methods of Mathematics: Calculus, Statistics, and Combinatorics

**Grader & T.A.** Pohang
POSTECH 2016 – 2018

- (2018 Spring) Differential Manifolds and Lie groups (Graduate course)
- (2017 Fall) Modern Algebra II
- (2017 Spring) Calculus
- (2016 Fall) Applied Linear Algebra (Undergraduate T.A.)

**Tutoring** Pohang
POSTECH 2014 – 2015

- (2015 Spring) Calculus
- (2015 Spring) Modern Algebra I
- (2014 Fall) Analysis II
- (2014 Spring) Analysis I

## Outreach

**KIAS Winter School on Mathematics and AI** Jeongseon
Team Leader 2025 Winter

- Use machine learning to study Artin representations and abelian varieties over finite fields
- Team members: Dohoon Choi, Keunyoung Jeong, Jaehak Yi, Minseo Shin, Euntaek Lee, Taeyoung Kim

**Berkeley Math REU** Berkeley
Mentor 2025 Summer

- Diophantine equations on Fibonacci polynomials and ties in Chebyshev bias over finite fields
- Mentees: Graeme Bates, Ryan Jesubalan, Jane Lu, Hyewon Shim

**Directed Reading Program** Berkeley
Mentor 2023-present

- (2025 Fall) Group cohomology (Graeme Bates, Jane Lu)
- (2025 Spring, 2025 Fall) Modular forms (Dongho Kim)
- (2023 Fall) Elliptic curves (Jacob Martin)
- (2023 Spring) $p$-adic numbers (Lucas Xie)

**POSTECH Potential Development Camp for High School Students** POSTECH
Mentor 2015 Winter

- Mentoring high school students for college-level mathematics

## Talks

- Research Talks

- Aarhus Mathamtics & AI Workshop, Aarhus Institute of Advanced Studies, Aarhus, January 2026
  How Machines Learn Galois Groups
- Lean Together 2026, Online, January 2026
  Progress on the Sphere Packing Project (with Sidarth Hariharan and Chris Birkbeck)
- June E Huh Center for Mathematical Challenges, KIAS, Seoul, January 2026
  Positive Quasimodular forms and Linear Programming Bounds
- Number Theory Seminar, University of Wisconsin-Madison, Wisconsin, December 2025
  How Machines Learn Galois Groups
- EPFL, Lausanne, November 2025
  Sphere Packing, Sign Uncertainty Principle, Universal Optimality, and Positive Quasimodular forms
- Aarhus University, Aarhus, November 2025
  Sphere Packing, Sign Uncertainty Principle, Universal Optimality, and Positive Quasimodular forms
- Special Seminar, Carnegie Mellon University, Pittsburgh, November 2025
  How Machines Learn Galois Groups
- International Seminar on Automorphic Forms, Online, April 2025
  Algebraic proof of modular form inequalities for optimal sphere packings
- RTG seminar, UC Berkeley, Berkeley, February 2025
  Algebraic proof of modular form inequalities for optimal sphere packings
- Algebra Discrete Math seminar, UC Davis, Davis, January 2025
  Algebraic proof of modular form inequalities for optimal sphere packings
- 6th EU/US Workshop on Automorphic Forms and Related Topics, Luminy, September 2024
  Algebraic proof of modular form inequalities for optimal sphere packings
- POSTECH Number Theory Seminar, POSTECH, Pohang, May 2024
  Algebraic proof of modular form inequalities for optimal sphere packings
- Student Number Theory Seminar, UC Berkeley, Berkeley, April 2024
  Algebraic proof of Viazovska's inequalities
- School of Mathematics, KIAS, Seoul, December 2023
  A new proof of Viazovska's modular form inequality and beyond
- International Conference on Machine Learning, Hawaii, US, July 2023
  HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- Center for Artificial Intelligence and Natural Sciences, KIAS, Seoul, June 2023
  HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- School of Computing, KAIST, Daejeon, June 2023
  HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- 1st FHE.org workshop, Trondheim, May 2022
  Encrypted Multinomial Logistic Regression Training with Softmax Approximation
- Workshop for Young Mathematicians in Korea, Online, January 2022
  Hitchhiker's guide to non-archimedean world
- Graduate student seminar, Sogang University, Seoul, July 2018
  Maass wave forms, quantum modular forms and Hecke operators
- Sungkyunkwan University, Seoul, June 2018
  Maass wave forms, quantum modular forms and Hecke operators
- NCTS-POSTECH Number Theory Workshop, NTU, Taiwan, December 2017
  Quantum modular forms and Hecke operators

- Expository Talks
  - Student Number Theory Seminar, Berkeley, March 2026
    Mathematics, AI, and Formalization
  - Harmonic Analysis Seminar, University of California Irvine, February 2026
    Mathematics, AI, and Formalization
  - Student Number Theory Seminar, Berkeley, August 2025
    How Do Automorphic Forms and Elliptic Curves Fly? (Survey on Murmuration)
  - Bruhat-Tits building seminar, Berkeley, February 2025
    Bruhat-Tits buildings for split groups / Moy-Prasad filtration and Local Langlands correspondence
  - Berkeley–Stanford Number Theory Learning Seminar, Berkeley, December 2024.
    Proof of irrationality of $L(2, \chi_{-3})$ and product of log values
  - Student Number Theory Seminar, Berkeley, October 2024
    Modular forms on $G_2$
  - Geometric class field theory learning seminar, Berkeley, Sep 2024
    Singular algebraic curves and de-normalization
  - Student Number Theory Seminar, Berkeley, March 2024
    Linear Programming Beyond Sphere Packing
  - Orbit methods and automorphic forms learning seminar, Berkeley, Oct 2023
    Gan–Gross–Prasad conjectures
  - Student Number Theory Seminar, Berkeley, Nov 2022
    Shimura correspondence and Waldspurger's formula
  - Instructional Workshop on Class Field Theory, KIAS, Seoul, January 2018
    Proof of the main theorem of local class field theory

## Languages

- Korean (native), English (fluent)
- Python (PyTorch, Numpy, Pandas), C/C++, LaTeX, SAGE Math, Lean, MATLAB, Haskell

## Miscelleneous (click the icons)

- Working as a reviewer for Mathematical Reviews (2022∼)
- GitHub blog on various topics
- Math Stackexchange & Math Overflow
- Speedcuber
- DJ (Techno, Trance, House)

(Last updated: March 25, 2026)